Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A ball of mass $2\ \text{kg}$ moving with a velocity of $8\ \text{ms}^{-1}$ collides head-on with another ball of mass $4\ \text{kg}$ moving with a velocity of $2\ \text{ms}^{-1}$ moving in the same direction. The collision is elastic and the coefficient restitution is $e = 0.5$.
(a) Find the velocities of the balls after the collision.
(b) Calculate the loss of kinetic energy due to collision.

(A) $5\ \text{ms}^{-1}$, 18 J

(B) $10\ \text{ms}^{-1}$, 28 J

(C) $25\ \text{ms}^{-1}$, 10 J

(D) $15\ \text{ms}^{-1}$, 15 J

Q.2

A ball of mass m moving horizontally at a speed v collides with the bob of a simple pendulum at rest. The mass of the bob is also m. If the collision is perfectly inelastic, the height to which the two balls rise after the collision will be given by

(A) $\frac{v^2}{g}$

(B) $\frac{v^2}{2g}$

(C) $\frac{v^2}{4g}$

(D) $\frac{v^2}{8g}$

Q.3

A block of wood of mass M is suspended by means of a thread. A bullet of mass m is fired horizontally into the block with a velocity v. As a result of the impact, the bullet is embedded in the block. The block will rise to vertical height given by

(A) $\frac{1}{2g}\left(\frac{mv}{M+m}\right)^2$

(B) $\frac{1}{2g}\left(\frac{mv}{M-m}\right)^2$

(C) $\frac{1}{2g}\frac{mv^2}{(M+m)}$

(D) $\frac{1}{2g}\frac{mv^2}{(M-m)}$

Q.4

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A body falls from a height h on a horizontal surface and rebounds. Then it falls again, and again rebounds and so on. If the restitution coefficient is $\frac{1}{3}$, the total distance covered by the body before it comes to rest is

(A) $\frac{h}{4}$

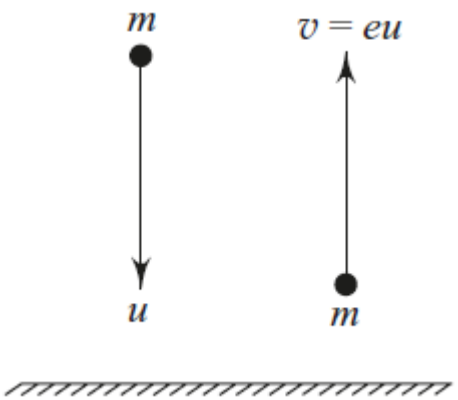
(B) $\frac{5h}{4}$

(C) $2h$

(D) $3h$

Q.5

A body is dropped from rest from a height $h = 5.0\ \text{m}$. After rebounding twice from a horizontal floor, to what height will it rise if the coefficient of restitution is 0.8 ?



(A) 2.03 m

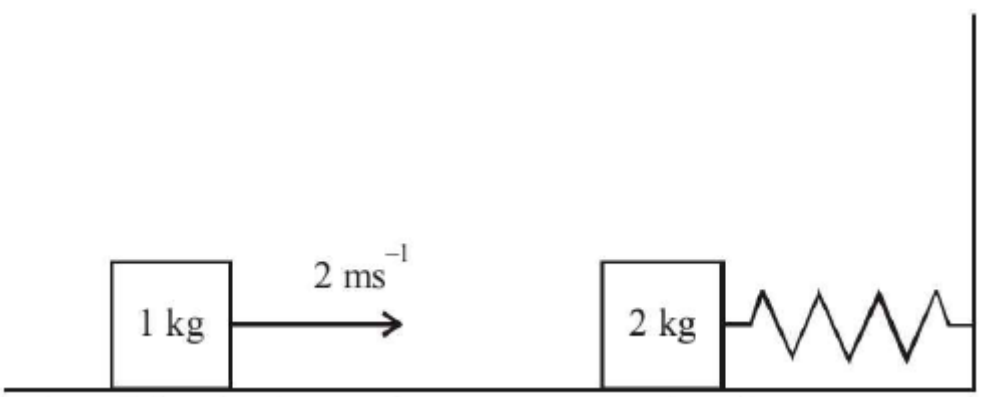
(B) 2.08 ms

(C) 2.05 m

(D) 1.95 m

Q.6

A spring-block system is resting on a frictionless floor as shown in the figure. The spring constant is 2.0Nm^{-1} and the mass of the block is $2.0\ \text{kg}$. Ignore the mass of the spring. Initially the spring is in an unstretched condition. Another block of mass $1.0\ \text{kg}$ moving with a speed of $2.0\ \text{ms}^{-1}$ collides elastically with the first block. The collision is such that the $2.0\ \text{kg}$ block does not hit the wall. The distance, in metres, between the two blocks when the spring returns to its unstretched position for the first time after the collision is ____



(A) 2.1

(B) 2.09

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(C) 3.1

(D) 3.09

Q.7

A body of mass $2\ \text{kg}$ moving with a velocity $(\hat{\text{i}}+2\hat{\text{j}}-3\hat{\text{k}})\text{ms}^{-1}$ collides with another body of mass $3\ \text{kg}$ moving with a velocity $(2\hat{\text{i}}+\hat{\text{j}}+\hat{\text{k}})\text{ms}^{-1}$.
If they stick together, the velocity in ms^{-1} of the composite body is

(A) $\frac{1}{5}(8\hat{\text{i}}+7\hat{\text{j}}-3\hat{\text{k}})$

(B) $\frac{1}{5}(-4\hat{\text{i}}+\hat{\text{j}}-3\hat{\text{k}})$

(C) $\frac{1}{5}(8\hat{\text{i}}+\hat{\text{j}}-\hat{\text{k}})$

(D) $\frac{1}{5}(-4\hat{\text{i}}+7\hat{\text{j}}-3\hat{\text{k}})$

Q.8

A ball hits a floor and rebounds after an inelastic collision. In this case

(A) the momentum of the ball just after the collision is the same as that just before the collision

(B) the mechanical energy of the ball remains the same is the collision

(C) the total momentum of the ball and the earth is conserved

(D) the total energy of the ball and the earth is conserved

Q.9

A metal ball of mass $2\ \text{kg}$ moving with speed of $36\ \text{km/h}$ has a head-on collision with a stationary ball of mass $3\ \text{kg}$. If after collision, both the balls move together, then the loss in kinetic energy due to collision is

(A) $40\ \text{J}$

(B) $60\ \text{J}$

(C) $100\ \text{J}$

(D) $140\ \text{J}$

Q.10

A bullet of mass $20\ \text{g}$ traveling horizontally with a speed of $500\ \text{m/s}$ passes through a wooden block of mass $8.0\ \text{kg}$ initially at rest on a level surface. The bullet emerges with a speed of $100\ \text{m/s}$ and the block slides $20\ \text{cm}$ on the surface before coming to rest. The coefficient of friction between the block and the surface is $(g = 10\ \text{ms}^{-2})$



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(A) 0.4

(B) 0.25

(C) 0.2

(D) 0.16

Q.11

A ball of mass m moving with a speed u undergoes a head-on elastic collision with a ball of mass nm initially at rest. The fraction of initial energy transferred to the heavier ball is

(A) $\frac{n}{1+n}$

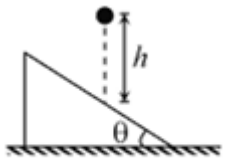
(B) $\frac{n}{(1+n)^2}$

(C) $\frac{2n}{(1+n)^2}$

(D) $\frac{4n}{(1+n)^2}$

Q.12

A ball after falling through a distance h collides with an inclined plane of inclination θ as shown. It moves horizontally after the impact. The co-efficient of restitution between inclined plane and ball is



(A) 1

(B) $\tan^2\theta$

(C) $\cot^2\theta$

(D) $\sin^2\theta$

Q.13

A glass marble dropped from a certain height above the horizontal surface reaches the surface in time t and then continues to bounce up and down. The time in which the marble finally comes to rest is (coefficient of restitution is e)

(A) $e^n t$

(B) $e^2 t$

(C) $t\left[\frac{1+e}{1-e}\right]$

(D) $t\left[\frac{1-e}{1+e}\right]$

Q.14

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A plate remains in equilibrium in air when n bullets are fired per second on it. The mass of each bullet is m and it strikes the plate with speed v. The coefficient of restitution is $e = 0.5$. The mass of the plate is

(A) $M = \frac{3mnv}{2g}$

(B) $M = \frac{mnv}{g}$

(C) $M = \frac{2mnv}{3g}$

(D) $M = \frac{2mnv}{g}$

Q.15

A small ball thrown at an initial velocity u directed at an angle $\theta = 37^\circ$ above the horizontal collides inelastically $(e = 1/4)$ with a vertical massive wall moving with a uniform horizontal velocity $u/5$ towards ball. After collision with the wall, the ball returns to the point from where it was thrown. Neglect friction between ball and wall. The time from beginning of motion of the ball till the moment of its impact with the wall is $(\tan 37^\circ = 3/4)$

(A) $\frac{3u}{5g}$

(B) $\frac{18u}{25g}$

(C) $\frac{54u}{125g}$

(D) $\frac{54u}{25g}$

Answers & Solutions

Q.1 Answer:

5 ms^{-1}, 18 J

Solution:

(a) Given $m_1 = 2\text{ kg}, m_2 = 4\text{ kg}, u_1 = 8\text{ ms}^{-1}, u_2 = 2\text{ ms}^{-1}$ **and** $e = 0.5$

From conservation of momentum

$$\begin{aligned} m_1u_1 + m_2u_2 &= m_1v_1 + m_2v_2 \\ \Rightarrow \quad 2 \times 8 + 4 \times 2 &= 2v_1 + 4v_2 \\ \Rightarrow \quad 12 &= v_1 + 2v_2 \end{aligned}$$

Since $e = 0.5$**, we have**

$v_2 - v_1 = e(u_1 - u_2) = 0.5 \times (8 - 2) = 3$

Eliminating v_2 **from (i) and (ii) we get** $v_1 = 2\text{ ms}^{-1}$**. Using this in (i) or (ii), we get** $v_2 = 5\text{ ms}$

(b) Kinetic energy before collision is

$$\begin{aligned} K_i &= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 \\ &= \frac{1}{2} \times 2 \times (8)^2 + \frac{1}{2} \times 4 \times (2)^2 = 72\text{ J} \end{aligned}$$

Kinetic energy after collision is

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$$K_f = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$$
$$= \frac{1}{2} \times 2 \times (2)^2 + \frac{1}{2} \times 4 \times (5)^2 = 54 \text{ J}$$
$\therefore$ **Loss of K.E.** $= K_i - K_f = 72 - 54 = 18$ J

Q.2 Answer:

$\frac{v^2}{8g}$

Solution:

In a perfectly inelastic collision, two bodies stick together. After the collision, the speed of the ball and the bob (sticking together) is $v' = v/2$. The height to which they will rise is given by

$v' = \sqrt{2gh'}$

or $h' = \frac{v'^2}{2g} = \frac{v^2}{8g}$

Hence the correct choice is (d).

Q.3 Answer:

$\frac{1}{2g}\left(\frac{mv}{M+m}\right)^2$

Solution:

Let V be the velocity of the block with the bullet embedded in it at the time of impact. Then from the principle of conservation of momentum, we have

$mv = (M+m)V$

or $V = \frac{mv}{(M+m)}$

If the block, with the bullet embedded in it, rises to a vertical height h, then from the principle of conservation of energy, we have

$$\frac{1}{2}(M+m)V^2 = (M+m)gh$$
$$\text{or } V = \sqrt{2gh}$$

Using (ii) in (i), we get

$\sqrt{2gh} = \frac{mv}{(M+m)}$

Squaring this equation, we find that h is given correctly by choice (a).

Q.4 Answer:

$\frac{5h}{4}$

Solution:

Total distance $= h + 2e^2h + 2e^4h + \ldots\ldots$

$= h + 2e^2h\left(1 + e^2 + \ldots..\right)$

$= h + \frac{2e^2h}{1-e^2} = h\left(\frac{1+e^2}{1-e^2}\right)$

$= h\left(\frac{1+\left(\frac{1}{3}\right)^3}{1-\left(\frac{1}{3}\right)^2}\right) = \frac{5h}{4}$

Hence the correct choice is (b).

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Q.5 Answer:

2.05 m

Solution:

Speed of the body just before first impact with floor $= \sqrt{2gh}$

Speed just after first impact $= e\sqrt{2gh}$**. This is also the speed just before the second impact. Therefore, speed just after second impact** $= e^2\sqrt{2gh}$**. This is the initial speed for the upward motion of the body after the second impact,**

i.e. $u = e^2\sqrt{2gh}$**. Therefore, height attained after two impacts is**

$$h_2 = \frac{u^2}{2g} = \frac{1}{2g}\left(e^2\sqrt{2gh}\right)^2 = e^4 h$$
$$= (0.8)^4 \times 5 = 2.05 \text{ m}$$

Q.6 Answer:

2.09

Solution:

$$v_1 = \frac{(m_1-m_2)u_1}{m_1+m_2} + \frac{2m_2u_2}{m_1+m_2} = \frac{(1-2)2}{1+2} = \frac{-2}{3} \text{ ms}^{-1}$$
$$v_2 = \frac{(m_2-m_1)u_2}{m_1+m_2} + \frac{2m_1u_1}{m_1+m_2} = \frac{2\times1\times2}{1+2} = \frac{4}{3} \text{ ms}^{-1}$$

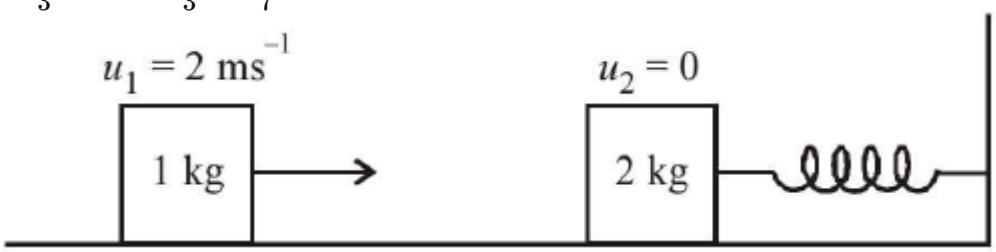
The time period of mass 2 kg **after attaining velocity is**

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{2}{2}} = 2\pi$$

Therefore the time taken to return the original position by 2 kg **mass is** πsec.

$\therefore$ **Distance between the two blocks**

$$= \frac{2}{3} \times \pi = \frac{2}{3} \times \frac{22}{7} = 2.09 \text{ m}$$



Q.7 Answer:

$\frac{1}{5}(8\hat{i} + 7\hat{j} - 3\hat{k})$

Solution:

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$\mathrm{p}_1 = 2\ \mathrm{kg}(\hat{\mathrm{i}} + 2\hat{\mathrm{j}} - 3\hat{\mathrm{k}})\mathrm{ms}^{-1} = (2\hat{\mathrm{i}} + 4\hat{\mathrm{j}} - 6\hat{\mathrm{k}})\mathrm{kgms}^{-1}$
$\mathrm{p}_2 = 3\ \mathrm{kg}(2\hat{\mathrm{i}} + \hat{\mathrm{j}} + \hat{\mathrm{k}})\mathrm{ms}^{-1} = (6\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 3\hat{\mathrm{k}})\mathrm{kgms}^{-1}$

Resultant momentum is

$$\begin{aligned}\mathrm{p} &= \mathrm{p}_1 + \mathrm{p}_2 \\ &= (2\hat{\mathrm{i}} + 4\hat{\mathrm{j}} - 6\hat{\mathrm{k}}) + (6\hat{\mathrm{i}} + 3\hat{\mathrm{j}} + 3\hat{\mathrm{k}}) \\ &= (8\hat{\mathrm{i}} + 7\hat{\mathrm{j}} - 3\hat{\mathrm{k}})\mathrm{kgms}^{-1}\end{aligned}$$

Total mass $(m) = 2 + 3 = 5$ **kg. Therefore, the velocity of composite body is**

$\mathrm{v} = \frac{\mathrm{P}}{m} = \frac{1}{5}(8\hat{\mathrm{i}} + 7\hat{\mathrm{j}} - 3\hat{\mathrm{k}})\mathrm{ms}^{-1}$

Hence the correct choice is (a).

Q.8 Answer:

the total momentum of the ball and the earth is conserved

Solution:

In an inelastic collision only momentum of the system may remain conserved. Some energy can be lost in the form of heat, sound etc.

Q.9 Answer:

60 J

Solution:

Loss in kinetic energy $= \frac{1}{2}\mu v_{\text{rel}}^2 = \frac{1}{2}\left(\frac{6}{5}\right)(10)^2 = 60\ \mathrm{J}$

Q.10 Answer:

0.25

Solution:

$m = 20g, M = 8kg, u = 500\ \mathrm{ms}^{-1}, v = 100\ \mathrm{ms}^{-1}$

By the law of conservation of momentum, $mu = mv + MV \Rightarrow V = 1\ \mathrm{ms}^{-1}$

$a = \mu g = \frac{V^2}{2s} \quad \therefore \quad \mu = 0.25$

Q.11 Answer:

$\frac{4n}{(1+n)^2}$

Solution:

$K_i = \frac{1}{2}mu^2$

By conservation of momentum $\quad mu = mv_1 + nmv_2$

$u = v_1 + nv_2 \quad \ldots$ **(i)**

$v_2 - v_1 = u_1 - u_2$

$\therefore \quad v_2 - v_1 = u \cdots$ **(ii)**

(i) + (ii) $\Rightarrow \quad 2u = (n+1)v_2 \quad v_2 = \frac{2u}{n+1}$

Kinetic energy transferred $\quad K_t = \frac{1}{2}(nm)v_2^2, \quad K_t = \frac{2nmu^2}{(n+1)^2}, \quad \frac{K_t}{K_i} = \frac{4n}{(n+1)^2}$

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Q.12 Answer:

$\tan^2\theta$

Solution:

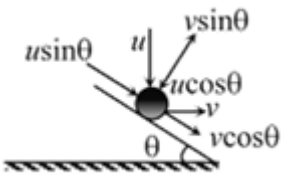
Impact takes place along the normal to the inclined plane

$\therefore \quad u\sin\theta = v\cos\theta$

$v = u\tan\theta ...(i)$

$e = \frac{v\sin\theta}{u\cos\theta} = \frac{u\tan\theta\cdot\sin\theta}{u\cos\theta}$

$e = \tan^2\theta$



Q.13 Answer:

$t\left[\frac{1+e}{1-e}\right]$

Solution:

Let height be h, so $t = \sqrt{\frac{2h}{g}}$

and $v = \sqrt{2gh} = gt$

Now $T = t + \frac{2ev}{g} + \frac{2e^2v}{g} + \ldots\ldots$

$= t + \frac{2ev}{g}\left(1 + e + e^2 + \ldots\right)$

$= t + \frac{2gt}{g}\left(\frac{e}{1-e}\right) = t\left(\frac{1+e}{1-e}\right)$

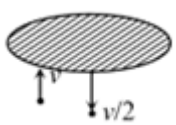
Q.14 Answer:

$M = \frac{3mnv}{2g}$

Solution:

The change in momentum of one bullet = final momentum − initial momentum

$= m\left(\frac{v}{2}\right) - (-mv) = \frac{3}{2}mv$



Also, given that n bullet strikes per second.

Force exerted by the bullets $= \frac{3}{2}mnv$

Since, the plate remains in equilibrium

$\therefore$ Force exerted by the bullets = weight of the plate

$\frac{3}{2}mnv = Mg$ **or** $M = \frac{3mnv}{2g}$ **(where M = mass of the plate)**

Q.15 Answer:

$\frac{54u}{125g}$

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Solution:

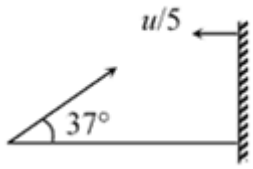
Let the ball collides with the wall after time t. Let velocity of ball after collision is v.

$\frac{-v-\left(-\frac{u}{5}\right)}{-\frac{u}{5}-u\cos 37} = \frac{1}{4}, \quad -v+\frac{u}{5} = -\frac{u}{4}$

$v = \frac{u}{5}+\frac{u}{4} = \frac{9u}{20}$

Also, $(u\cos 37)t = \frac{9u}{20}(T-t)$

$\frac{4ut}{5} = \frac{9u}{20}\left(\frac{2u}{g}\frac{3}{5}-t\right) \quad \Rightarrow t = \frac{54u}{125g}$



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